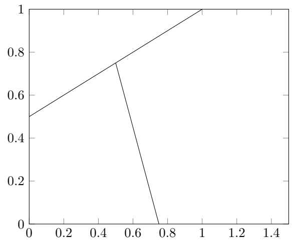
# 1 Mantle transport

## 1.1 Phase behavior

Three phases in total with 2 solid phases:

- Phase 1 can melt into fluid.
- Phase 2 can also melt into fluid.
- Different substances in fluid can react with each other.




Set up equations for component 1, which exhausts phase 1. Similarly, phase 2 is all component 2. Denote $c_{i,j}$ as component i in phase j.

Thus, the following equations can be derived:

Phase 1:

$$c_{1,1} = \rho_1 = \frac{M_{1,1}}{V_1} \qquad c_{2,1} = 0 = M_{2,1} \tag{1}$$

Note: $\rho_1 = \rho_1(p_1)$ only.

Phase 2:

$$c_{1,2} = 0 = M_{1,2} \qquad c_{2,2} = \rho_2 = \frac{M_{2,2}}{V_2} \tag{2}$$

Note: $\rho_2 = \rho_2(p_2)$ only.

Phase fluid:

$$c_f = c_{1,f} = \frac{M_{1,f}}{V_f}, \qquad , c_{2,f} = \frac{M_{2,f}}{V_f} \tag{3}$$

Additional relations and negligible diffusion:

$$\phi_f V_f \sim (\phi_1 + \phi_2) V_s, \qquad \phi_f D \nabla c_f \sim 0 \tag{4}$$

Transport variable $\bar{c}$ is defined as following:

$$\begin{aligned} \bar{c} &= \phi_f c_{1,f} + \phi_1 c_{1,1} + \phi_2 c_{1,2} \\ &= \phi_f c_f + \phi_1 c_s \end{aligned} \tag{5}$$

Note

$$\phi_f = \frac{V_f}{V}, \qquad , \phi_1 = \frac{V_1}{V}, \qquad , \phi_2 = \frac{V_2}{V} \tag{6}$$

$$\phi_s = \phi_1 + \phi_2 \tag{7}$$

$$\phi_f + \phi_1 + \phi_2 = \phi_f + \phi_s = 1 \tag{8}$$

Then

$$\begin{aligned} \bar{c} = \phi_f c_f + \phi_1 c_s &= \frac{V_f}{V} \frac{M_{1,f}}{V_f} + \frac{V_1}{V} \frac{M_{1,1}}{V_1} \\ &= (M_{1,f} + M_{1,1})/V \\ &= M_1/V \end{aligned} \tag{9}$$

The transport equation is now converted into:

$$\overline{c}_t + \nabla \cdot (\overline{cv} - \phi_f D \nabla c_f) = 0 \tag{10}$$

Two possible approaches:

(1) Linearize phase behavior:

$$\phi_f c_f = \kappa(\overline{H}, \overline{c})\overline{c} \approx \kappa \overline{c} \tag{11}$$

(2) Operator split by phase:

$$\begin{cases} (\phi_f c_f)_t + \nabla \cdot (\phi_f c_f v_f - \phi_f D \nabla c_f) = \Gamma \approx 0 \\ \qquad\qquad (\phi_1 c_s)_t + \nabla \cdot (\phi_1 c_s v_s) = -\Gamma \approx 0 \end{cases} \tag{12}$$

Flash calculation to approx $\Gamma$.

We pursue first approach (Eqn. 11). Note:

$$0 \leq \kappa \leq 1 \tag{13}$$

Since,

$$\phi_f c_f = \kappa \overline{c} = \kappa(\phi_f c_f + \phi_1 c_s)$$

moreover,

$$\overline{c} = \phi_f c_f + \phi_1 c_s$$

$\Rightarrow$

$$\phi_1 c_s = \overline{c} - \phi_f c_f = (1 - \kappa)\overline{c} \tag{14}$$

now we have

$$\begin{aligned} \overline{cv} &= \phi_f c_f v_f + \phi_1 c_s v_s + \phi_2 \cdot 0 v_s \\ &= (\kappa v_f + (1 - \kappa) v_s)\overline{c} \end{aligned}$$

$\Rightarrow$

$$\overline{cv} = v_e \overline{c} \tag{15}$$

$$v_e = K v_f + (1 - K) v_s \tag{16}$$

So previous equation becames:

$$\overline{c}_t + \nabla \cdot (v_e \overline{c} - \phi_f D \nabla c_f) = 0 \tag{17}$$

Also

$$\begin{aligned} \phi_f D \nabla c_f &= \phi_f D \nabla (\frac{\kappa \overline{c}}{\phi_f}) \\ &= \kappa D \nabla \overline{c} + \overline{c} D (\nabla \kappa - \kappa \nabla ln(\phi)) \end{aligned} \tag{18}$$

Thus transport equation (17) becomes:

$$\overline{c}_t + \nabla \cdot (v_e^* \overline{c} - \kappa D \nabla \overline{c}) = 0 \tag{19}$$

where

$$\begin{aligned} v_e^* &= v_e - D(\nabla K - K \nabla ln(\phi)) \\ &= \kappa v_f + (1 - K) v_s - D(\nabla \kappa - \kappa \nabla ln(\phi)) \end{aligned} \tag{20}$$

Then phase calculation for new $c_f$, $c_s$, $\phi_f$, $\phi_1$.

Questions:

- 1. Is approach (11) better than (12)?
- 2. Can we approximate the nonlinearity in $\kappa(\overline{c})$ to solve (19) more accurately?
- 3. Can we get $\kappa(\overline{c})$ itself? Then solve (19) without needing flash calculatio.

## 1.2 Thermodynamics

Let

$$E = \overline{\rho e} = \text{internal energy/volume} \tag{21}$$

Thus the energy conservation equation is defined as:

$$\frac{\partial}{\partial t}\overline{\rho e} + \nabla \cdot (\overline{\rho e v} - \overline{K}\nabla T) = 0 \tag{22}$$

Now suppose that

$$\begin{aligned} \phi_f \rho_f e_f &= \lambda(\overline{H}, \overline{C})\overline{\rho e} = \lambda \overline{\rho e} = \lambda E \\ &= \lambda(\phi_f \rho_f e_f + \phi_1 \rho_1 e_1 + \phi_2 \rho_2 e_2) \end{aligned} \tag{23}$$

where

$$\phi_1 \rho_1 e_1 + \phi_2 \rho_2 e_2 = (1 - \lambda)\overline{\rho e} \tag{24}$$

Thus the resulting energy conservation law is:

$$E_t + \nabla \cdot (v_e^E E - \overline{K}\nabla T) = 0 \tag{25}$$

which is derived similarly with what we have done above. where average velocity is defined as:

$$v_e^E = \lambda v_f + (1 - \lambda) v_s \tag{26}$$

Now

$$\overline{K} = \phi_f K_f + \phi_1 K_1 + \phi_2 K_2 \tag{27}$$

where

$$\nabla T(C, E) = \frac{\partial T}{\partial C}\nabla C + \frac{\partial T}{\partial E}\nabla E \tag{28}$$

At each time step, temperature distribution will be obtained directly from Marc's code, as well as the corresponding spactial gradient of temperature. The convergence of this method will be researched.

## 1.3 Summary

The following system will be dealt with:

$$\begin{cases} \bar{c}_t + \nabla \cdot (v_e^* \bar{c} - \kappa D \nabla \bar{c}) = 0 \\ E_t + \nabla \cdot (v_e^E E - \overline{K}\nabla T) = 0 \end{cases}$$